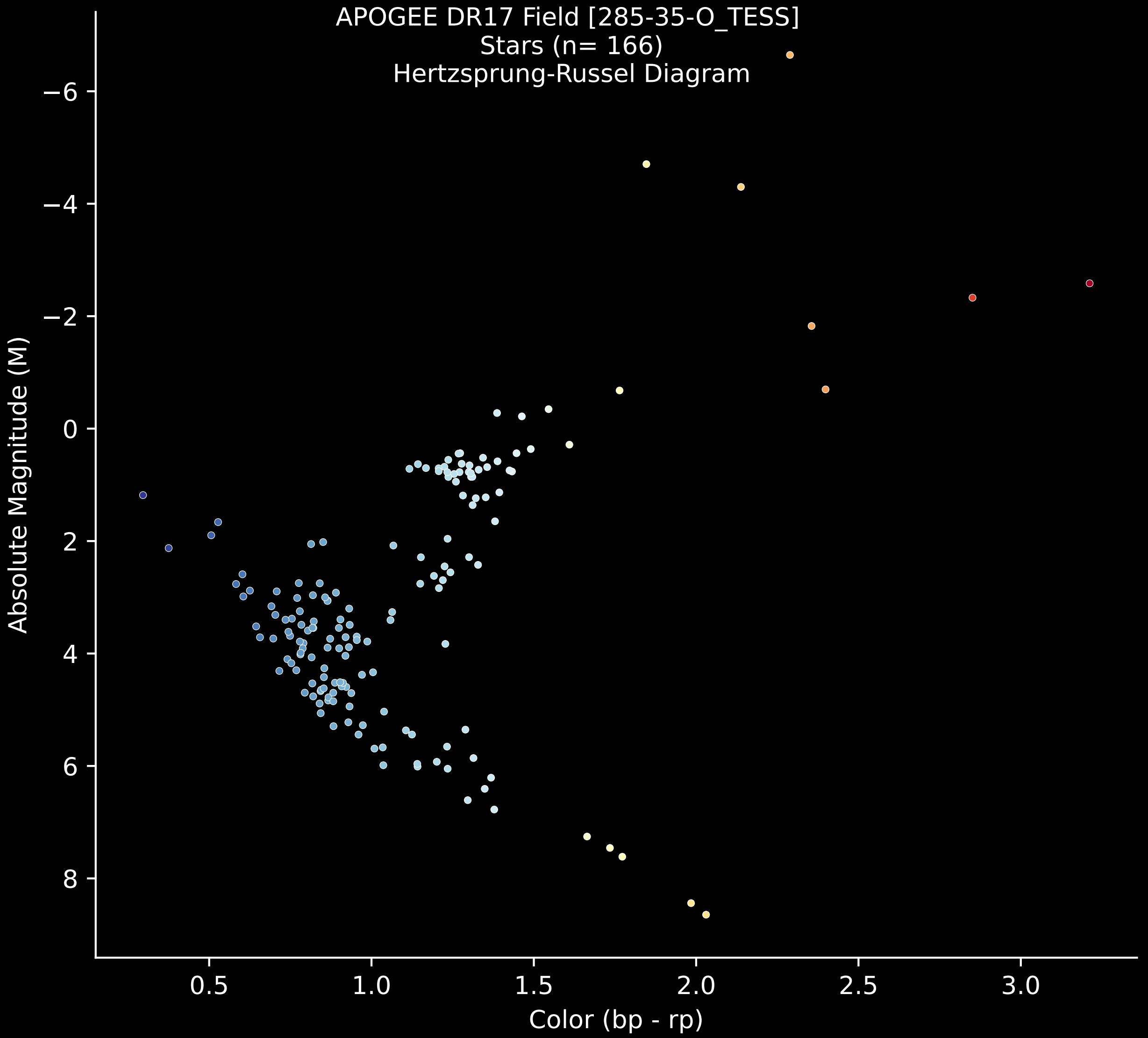
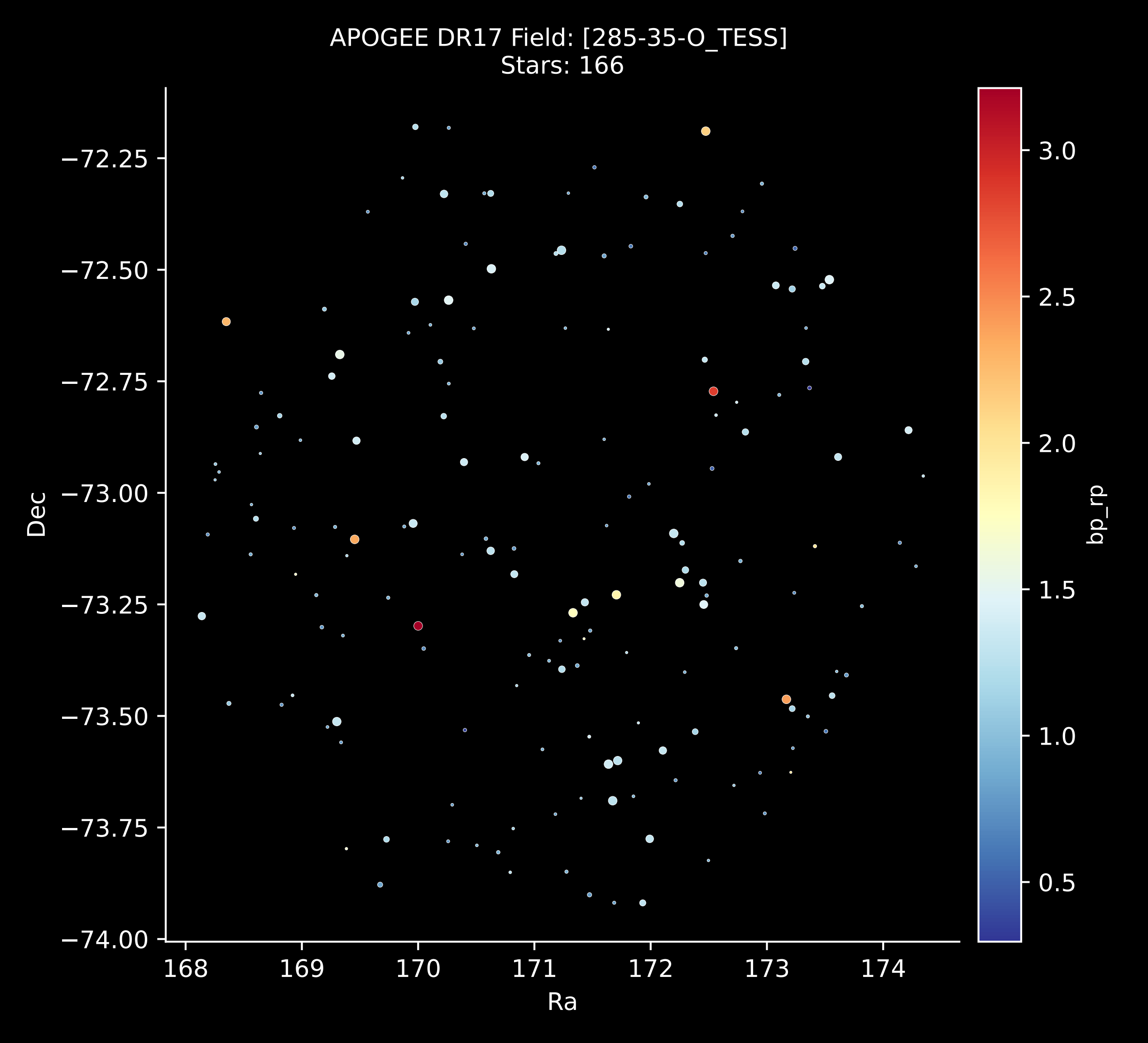
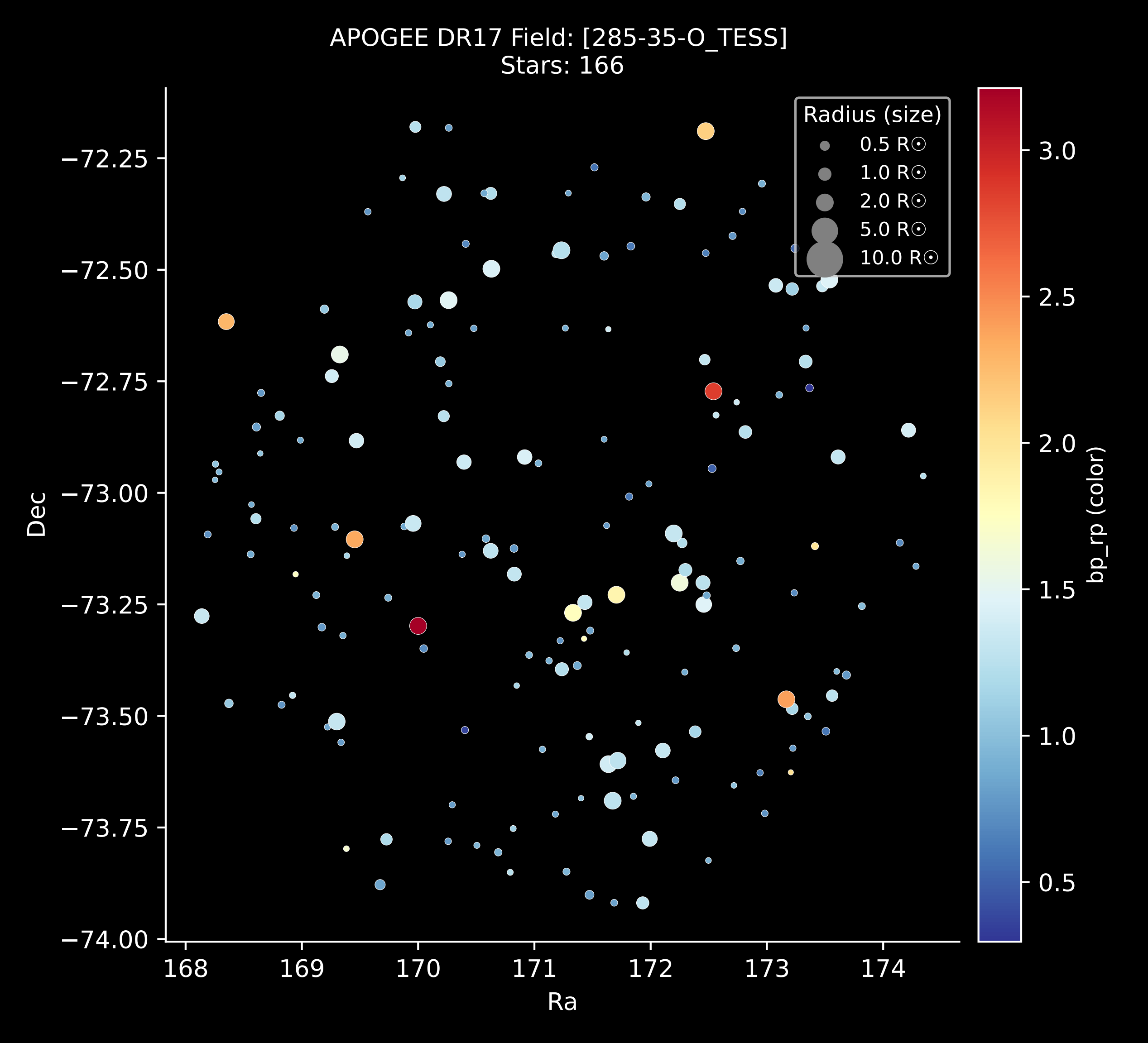
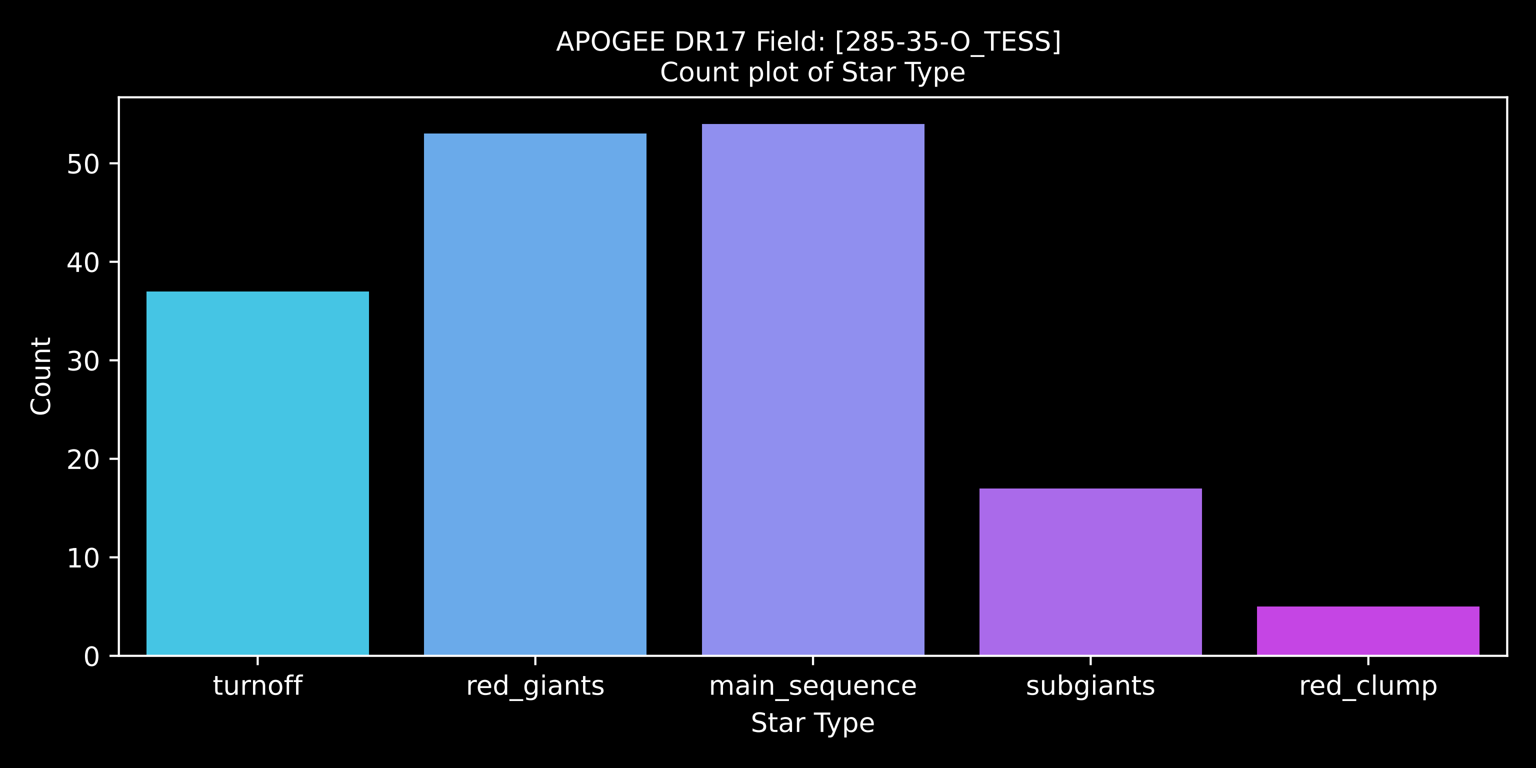
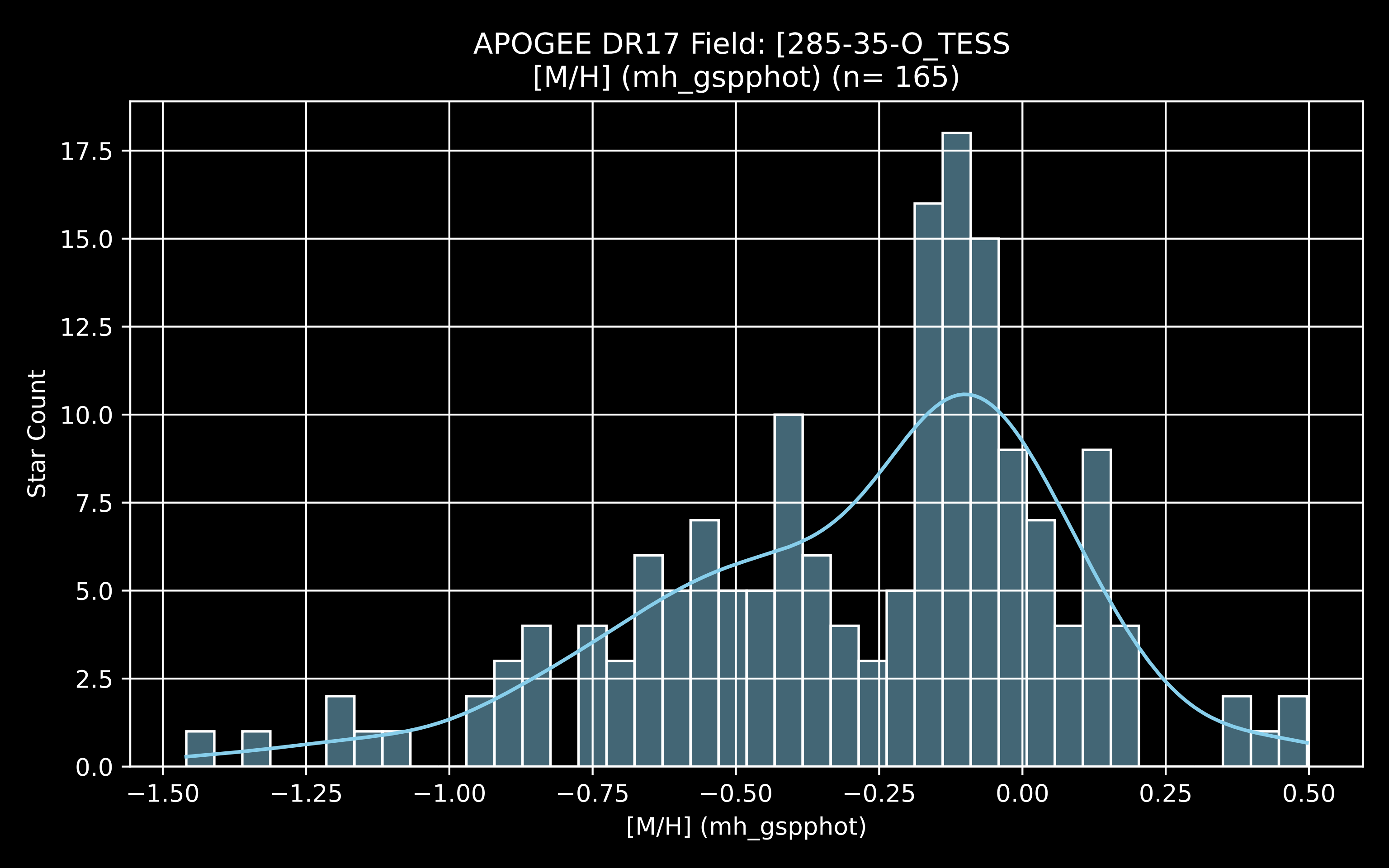


APOGEE DR17 Field [285-35-O_TESS]
Stars (n= 166)
Hertzsprung-Russel Diagram
Absolute Magnitude (M)
Color (bp - rp)
−6
−4
−2
0
2
4
6
8
0.5
1.0
1.5
2.0
2.5
3.0

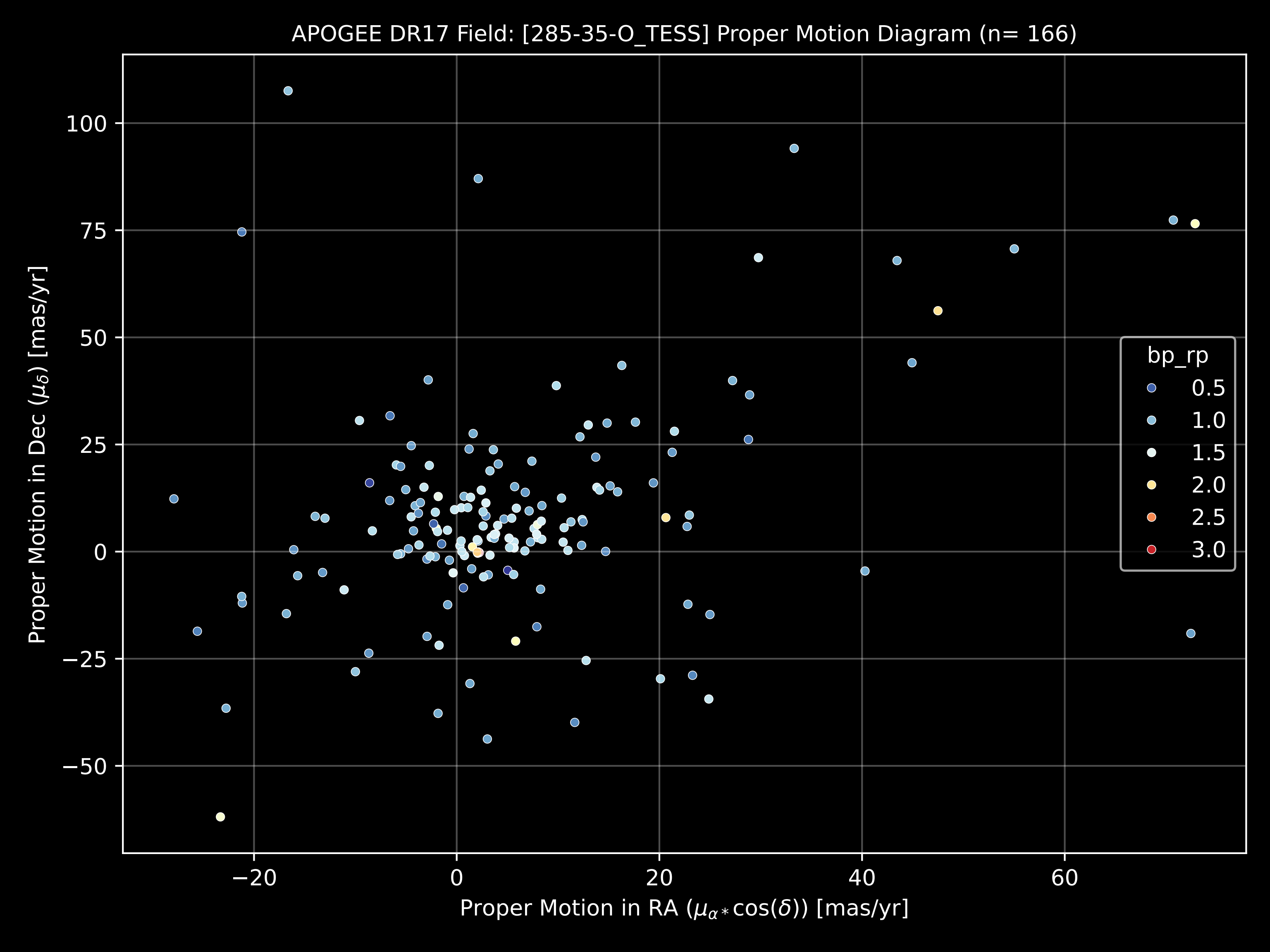

APOGEE DR17 Field: [285-35-O_TESS]
Stars: 166
Dec
Ra
bp_rp
−72.25
−72.50
−72.75
−73.00
−73.25
−73.50
−73.75
−74.00
168
169
170
171
172
173
174
3.0
2.5
2.0
1.5
1.0
0.5

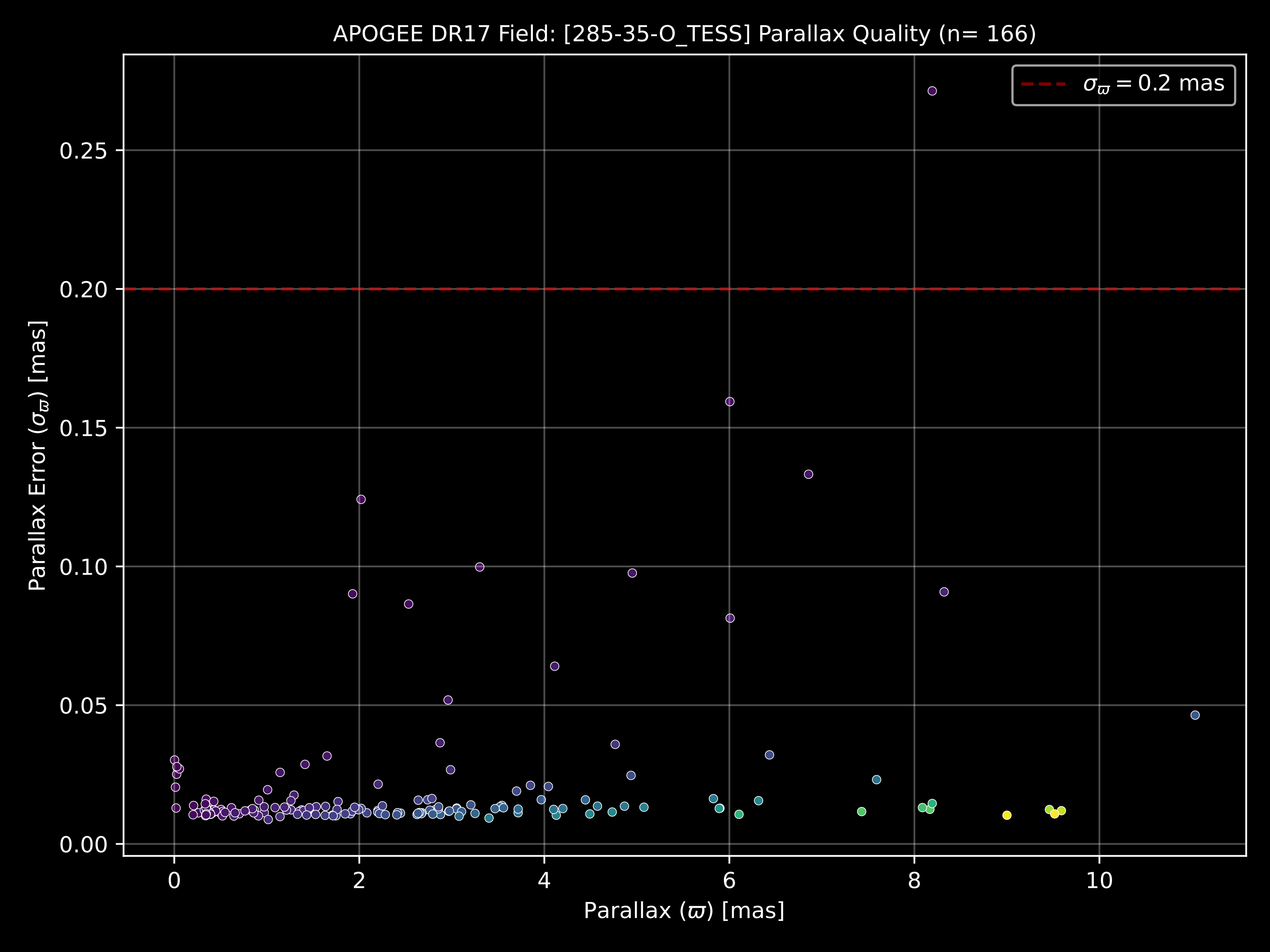

APOGEE DR17 Field: [285-35-O_TESS]
Stars: 166
Radius (size)
0.5 R☉
1.0 R☉
2.0 R☉
5.0 R☉
10.0 R☉
−72.25
−72.50
−72.75
−73.00
−73.25
−73.50
−73.75
−74.00
Dec
168
169
170
171
172
173
174
Ra
3.0
2.5
2.0
1.5
1.0
0.5
bp_rp (color)


APOGEE DR17 Field: [285-35-O_TESS]
Count plot of Star Type
Count
0
10
20
30
40
50
turnoff
red_giants
main_sequence
subgiants
red_clump
Star Type

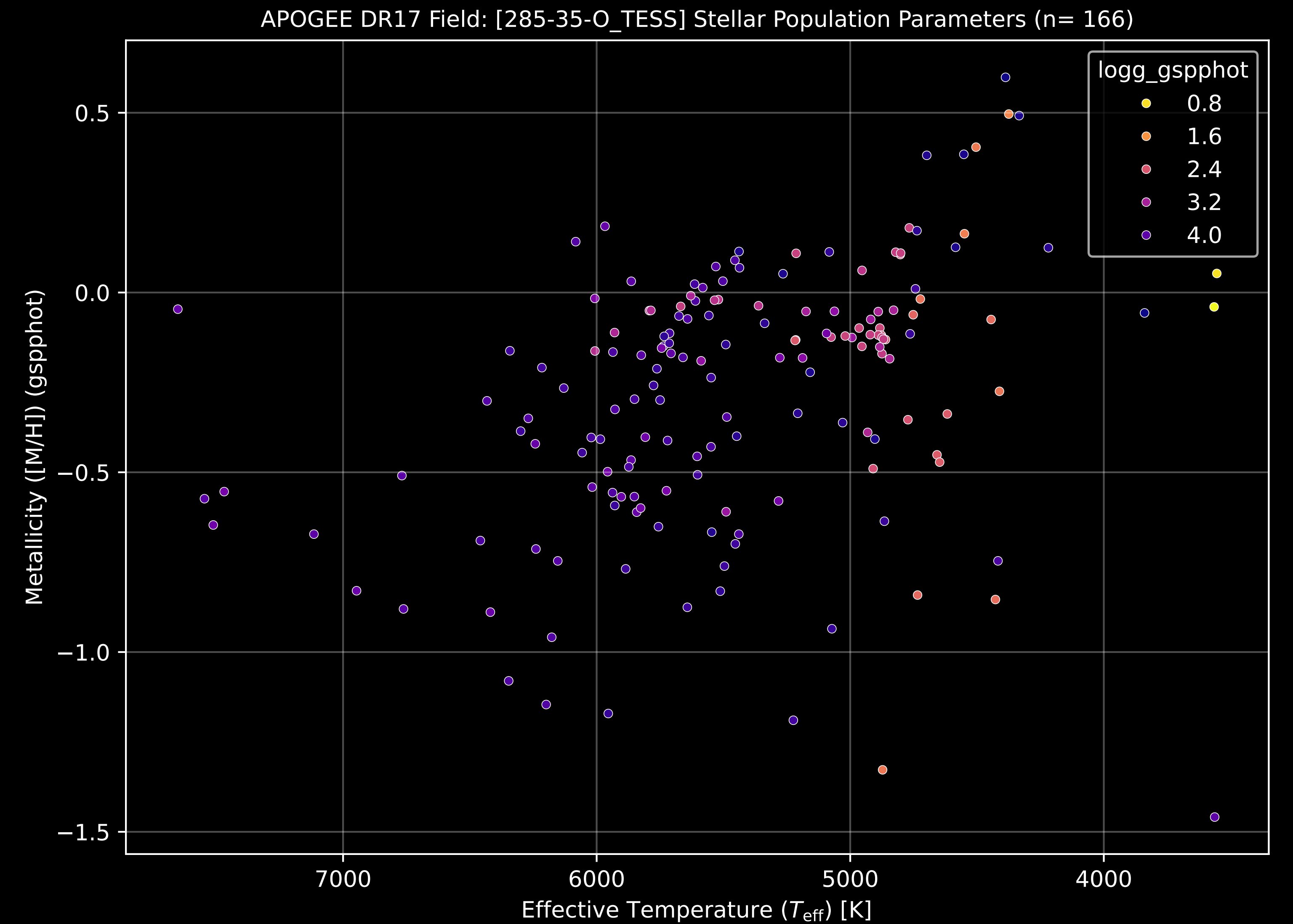


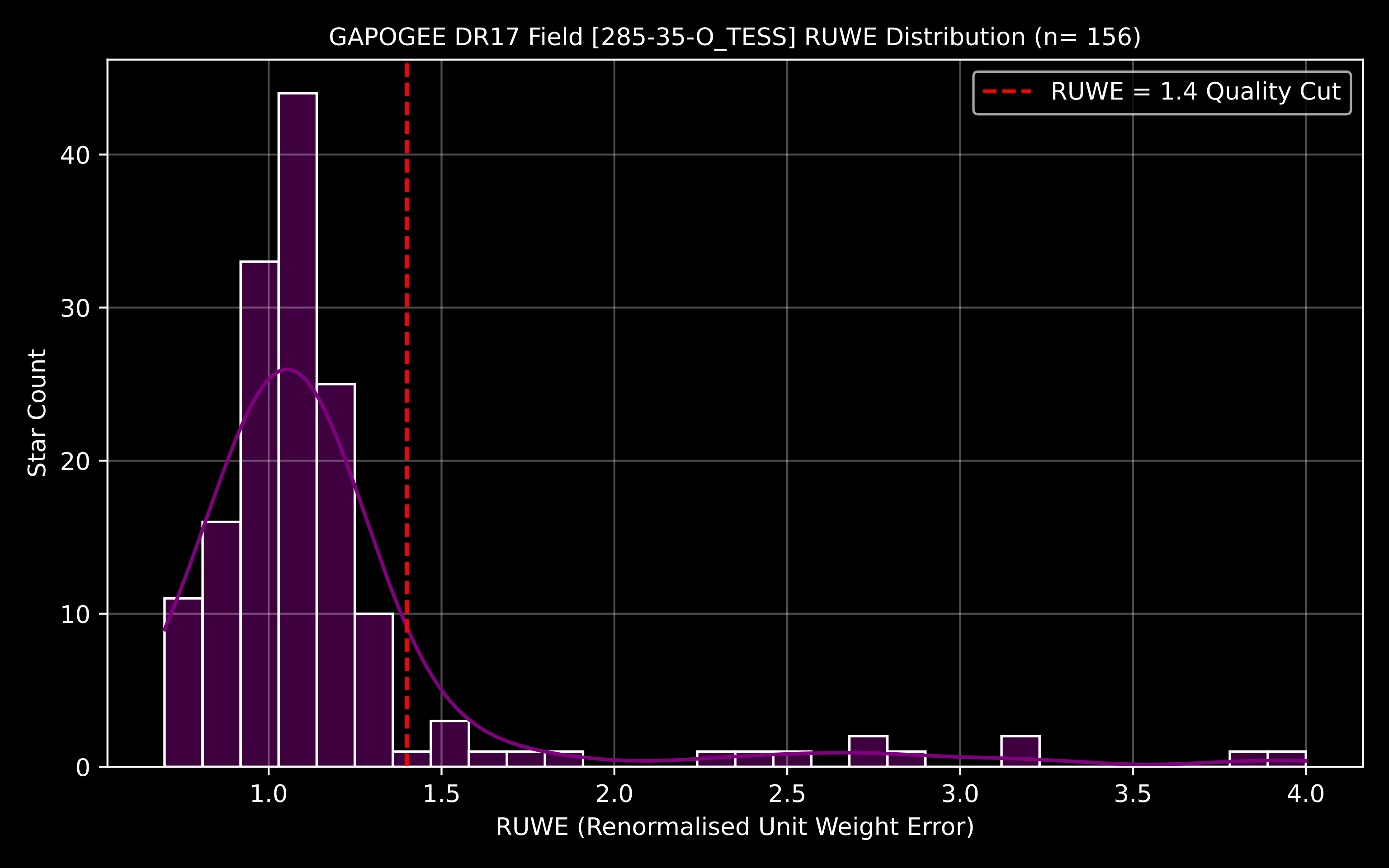
APOGEE DR17 Field: [285-35-O_TESS
[M/H] (mh_gspphot) (n= 165)
Star Count
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
[M/H] (mh_gspphot)

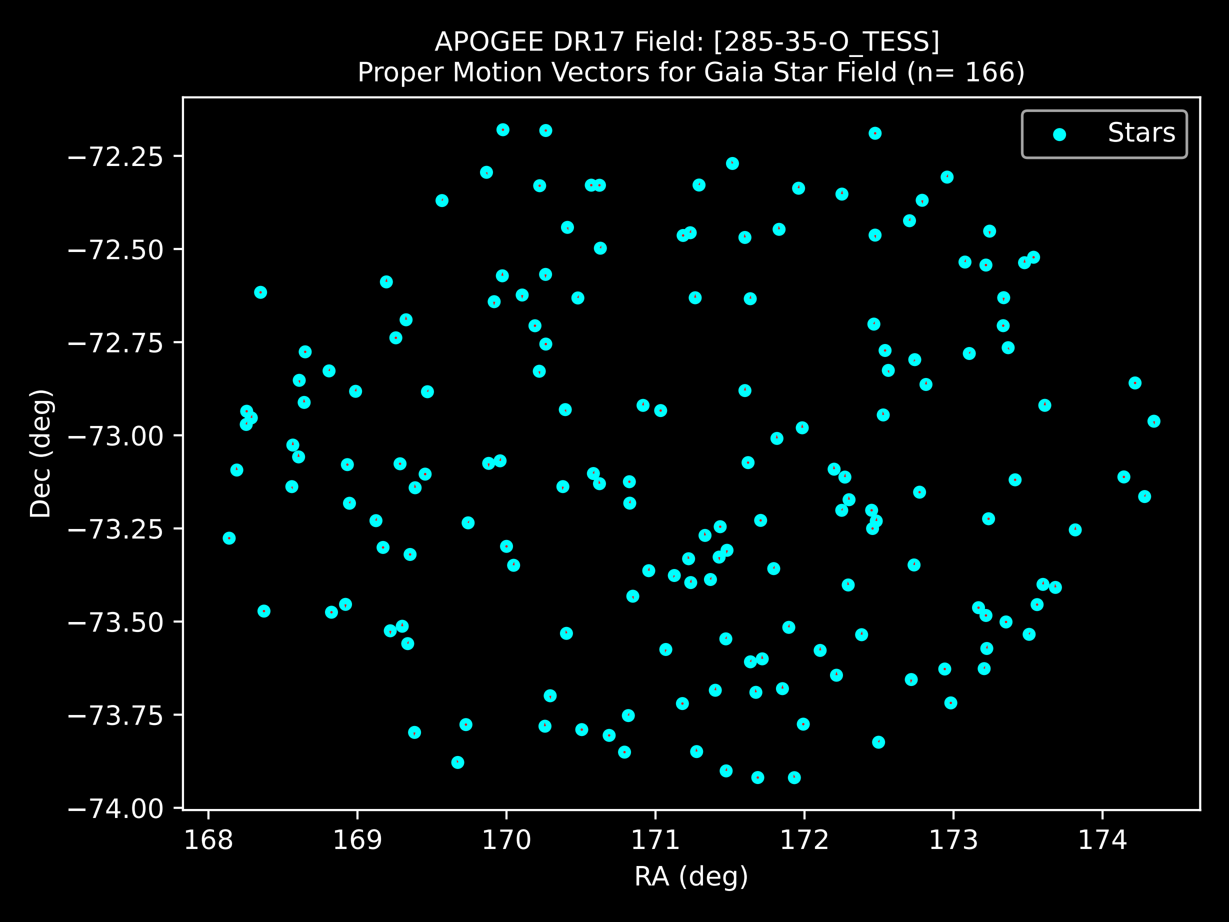

APOGEE DR17 Field: [285-35-O_TESS] Proper Motion Diagram (n= 166)
Proper Motion in Dec (μ_δ) [mas/yr]
Proper Motion in RA ($\mu_{\alpha*}\cos(\delta)$) [mas/yr]
100
75
50
25
0
−25
−50
−20
0
20
40
60
bp_rp
0.5
1.0
1.5
2.0
2.5
3.0



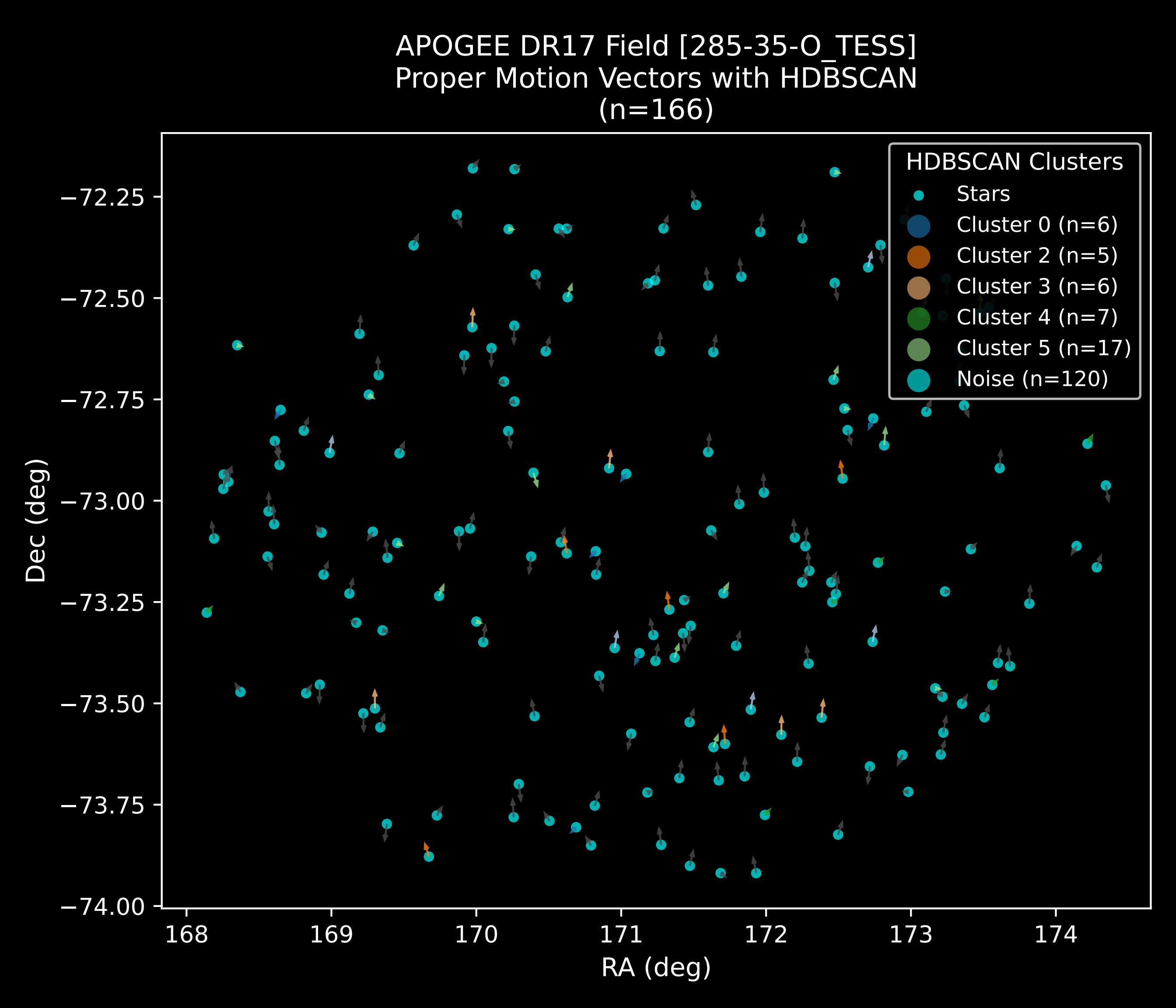
APOGEE DR17 Field: [285-35-O_TESS] Parallax Quality (n= 166)
$\sigma_\varpi = 0.2$ mas
Parallax Error (σ_ϖ) [mas]
Parallax (ϖ) [mas]
0.00
0.05
0.10
0.15
0.20
0.25
0
2
4
6
8
10


APOGEE DR17 Field: [285-35-O_TESS] Stellar Population Parameters (n= 166)
logg_gspphot
0.8
1.6
2.4
3.2
4.0
0.5
0.0
−0.5
−1.0
−1.5
Metallicity ([M/H]) (gspphot)
7000
6000
5000
4000
Effective Temperature ($T_{\rm eff}$) [K]

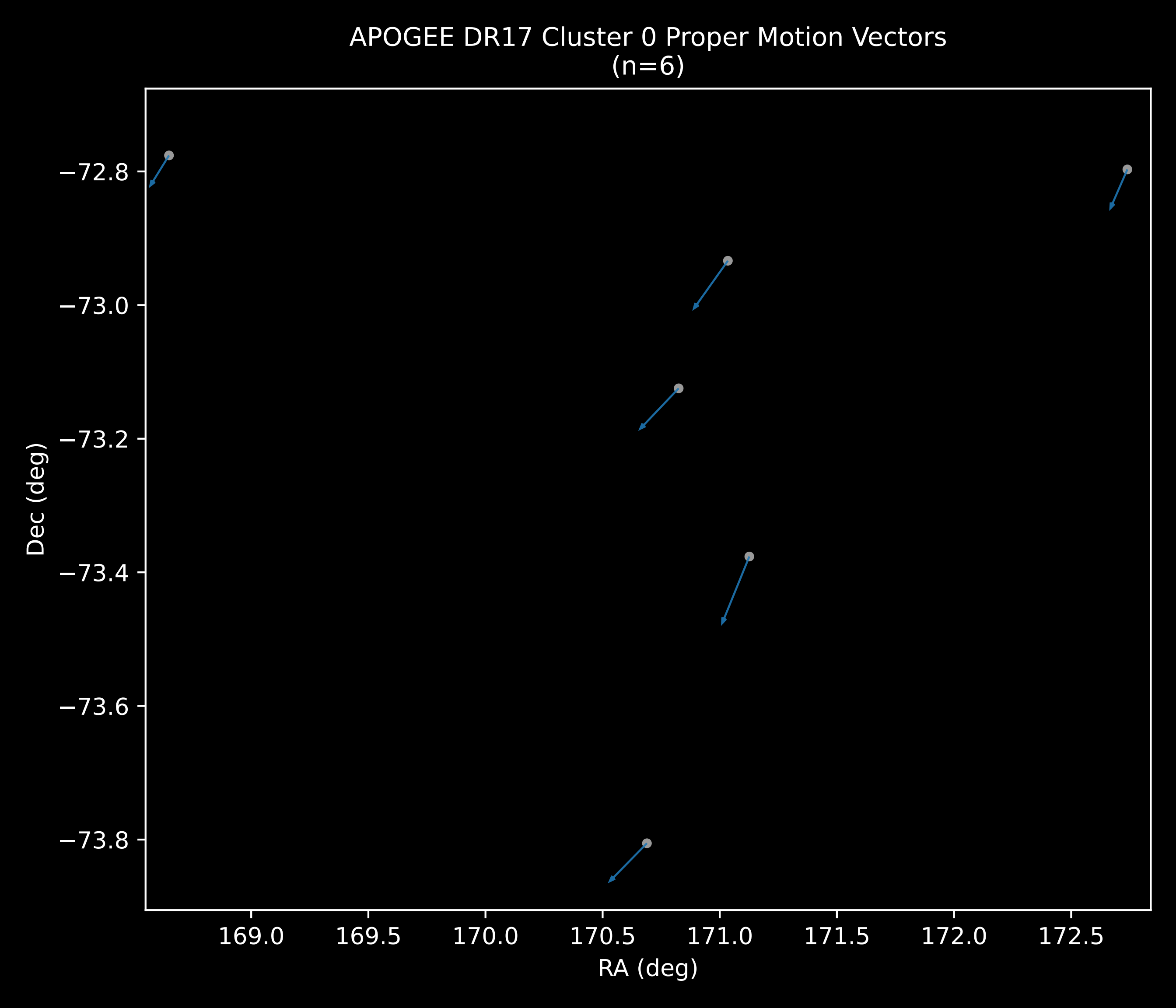

GAPOGEE DR17 Field [285-35-O_TESS] RUWE Distribution (n= 156)
RUWE = 1.4 Quality Cut
Star Count
0
10
20
30
40
1.0
1.5
2.0
2.5
3.0
3.5
4.0
RUWE (Renormalised Unit Weight Error)

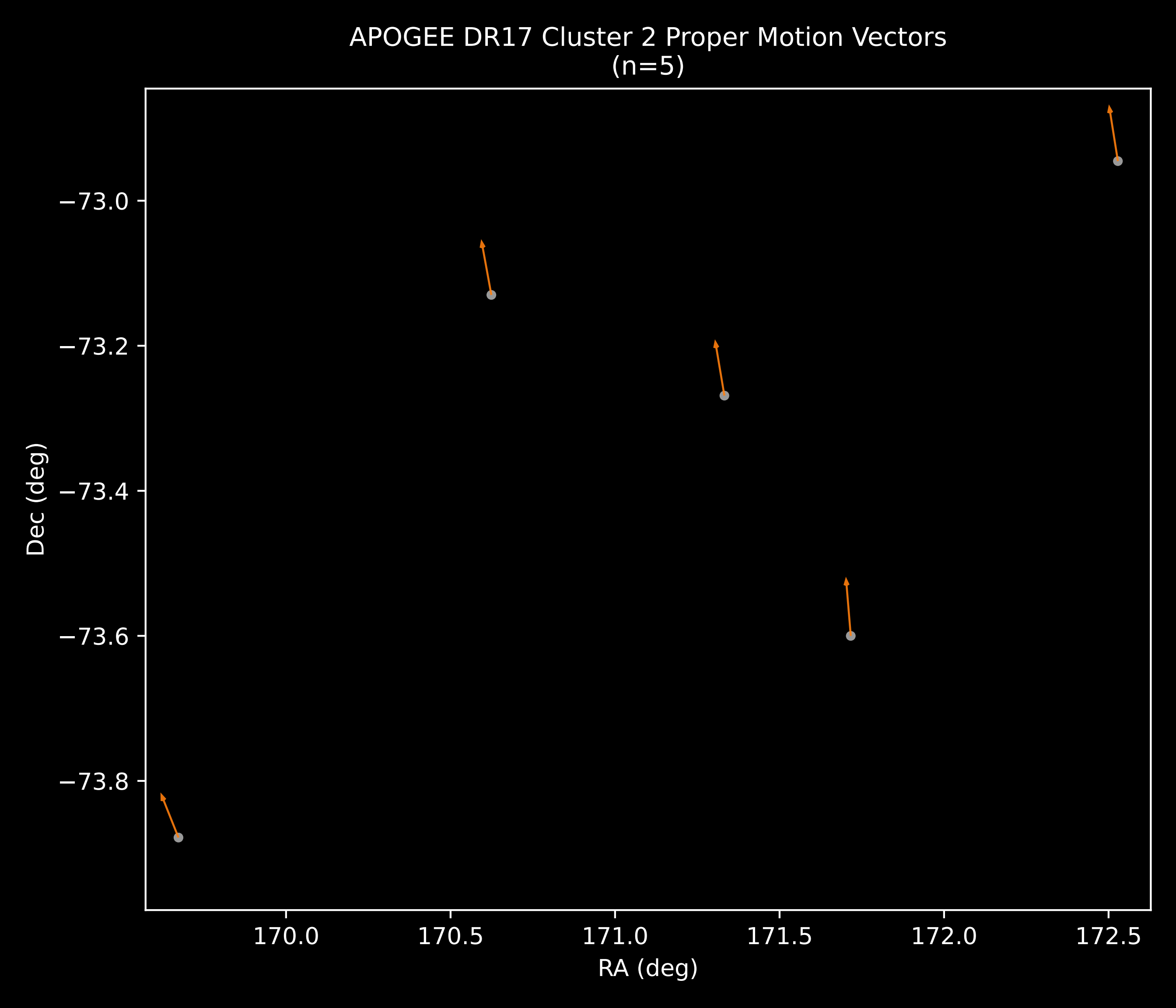


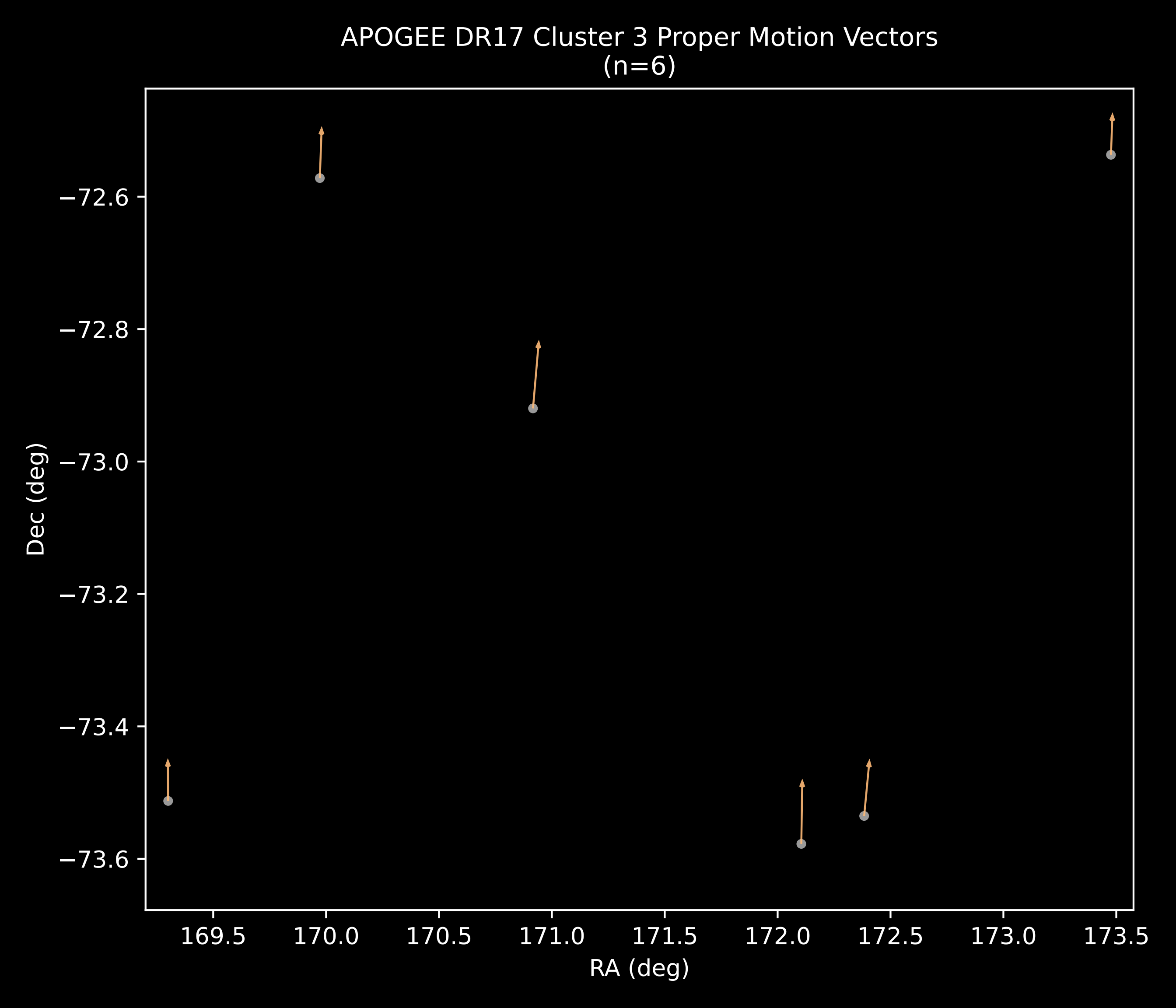
APOGEE DR17 Field: [285-35-O_TESS]
Proper Motion Vectors for Gaia Star Field (n= 166)
Stars
Dec (deg)
RA (deg)
−72.25
−72.50
−72.75
−73.00
−73.25
−73.50
−73.75
−74.00
168
169
170
171
172
173
174

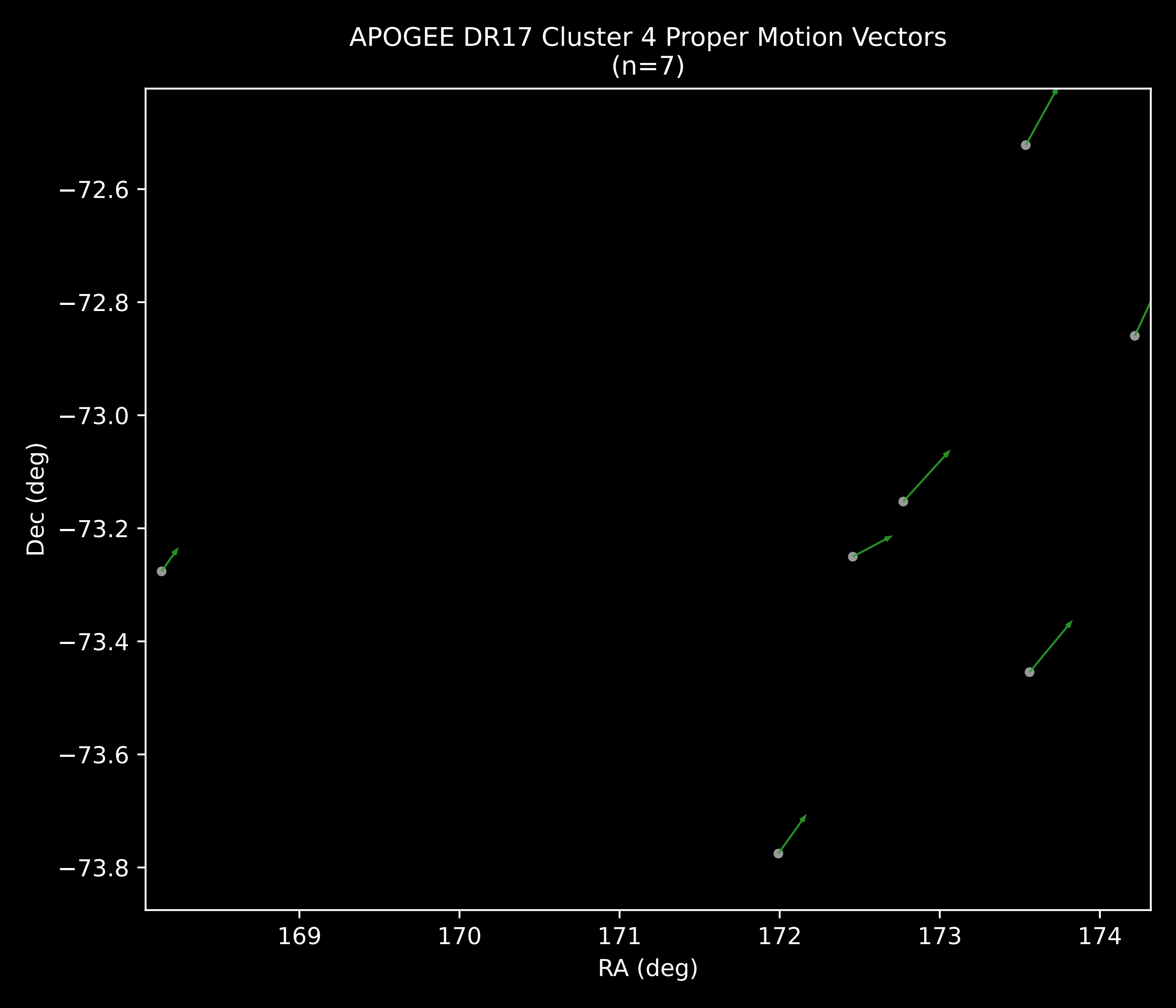


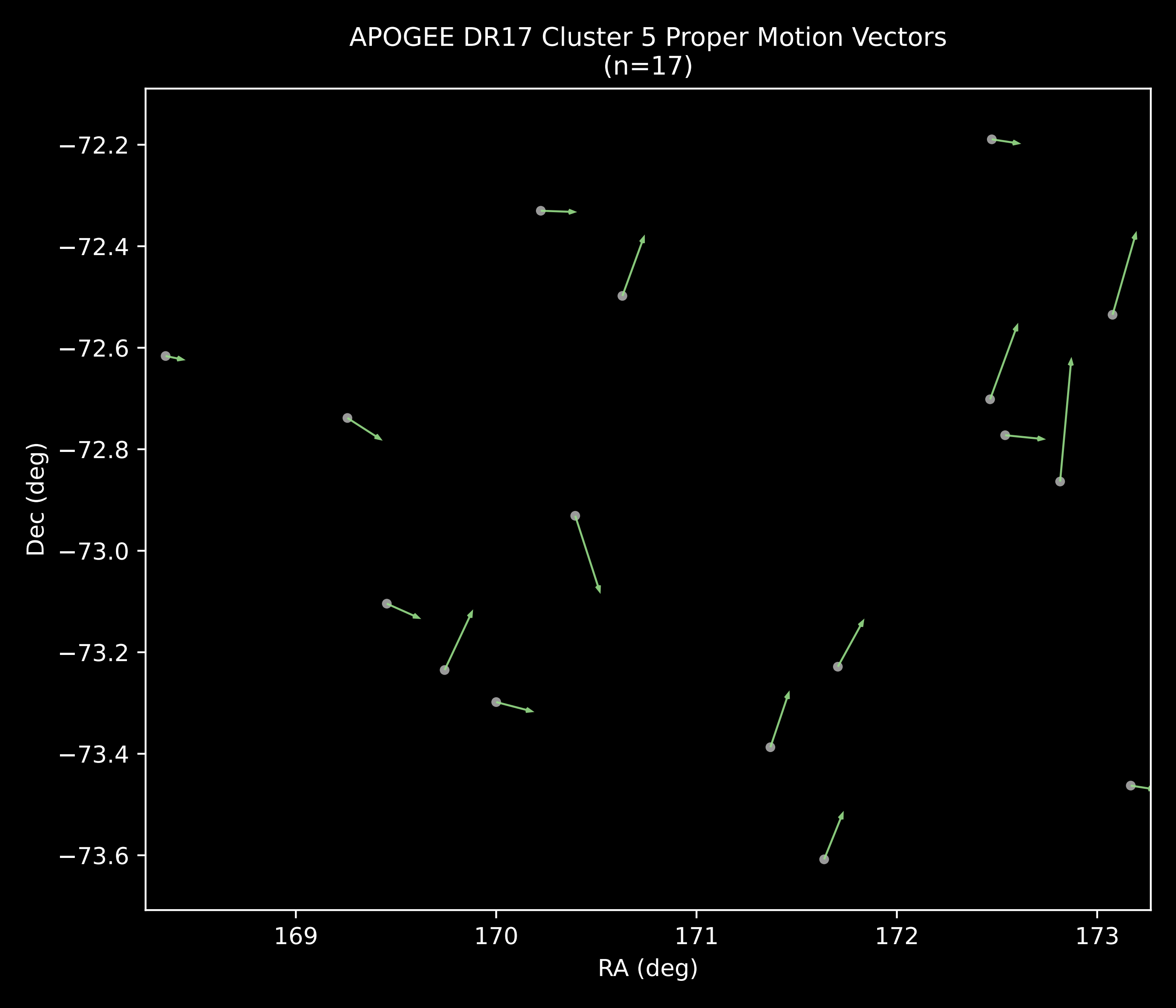
APOGEE DR17 Field [285-35-O_TESS]
Proper Motion Vectors with HDBSCAN
(n=166)
HDBSCAN Clusters
Stars
Cluster 0 (n=6)
Cluster 2 (n=5)
Cluster 3 (n=6)
Cluster 4 (n=7)
Cluster 5 (n=17)
Noise (n=120)
−72.25
−72.50
−72.75
−73.00
−73.25
−73.50
−73.75
−74.00
Dec (deg)
168
169
170
171
172
173
174
RA (deg)

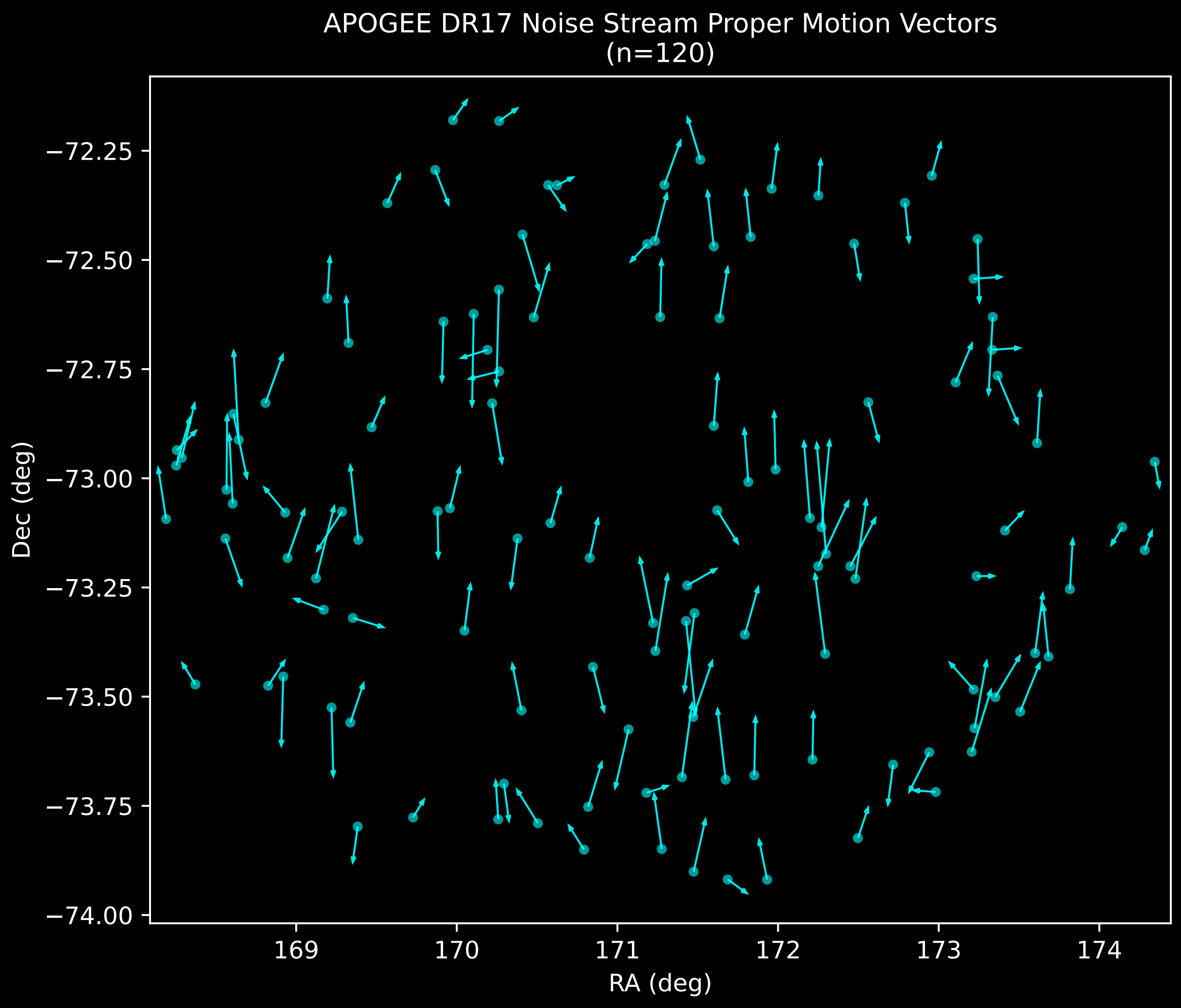

APOGEE DR17 Cluster 0 Proper Motion Vectors
(n=6)
Dec (deg)
RA (deg)
−72.8
−73.0
−73.2
−73.4
−73.6
−73.8
169.0
169.5
170.0
170.5
171.0
171.5
172.0
172.5


APOGEE DR17 Cluster 2 Proper Motion Vectors
(n=5)
Dec (deg)
−73.0
−73.2
−73.4
−73.6
−73.8
170.0
170.5
171.0
171.5
172.0
172.5
RA (deg)


APOGEE DR17 Cluster 3 Proper Motion Vectors
(n=6)
−72.6
−72.8
−73.0
−73.2
−73.4
−73.6
Dec (deg)
169.5
170.0
170.5
171.0
171.5
172.0
172.5
173.0
173.5
RA (deg)


APOGEE DR17 Cluster 4 Proper Motion Vectors
(n=7)
Dec (deg)
−72.6
−72.8
−73.0
−73.2
−73.4
−73.6
−73.8
169
170
171
172
173
174
RA (deg)


APOGEE DR17 Cluster 5 Proper Motion Vectors
(n=17)
Dec (deg)
−72.2
−72.4
−72.6
−72.8
−73.0
−73.2
−73.4
−73.6
RA (deg)
169
170
171
172
173


APOGEE DR17 Noise Stream Proper Motion Vectors
(n=120)
−72.25
−72.50
−72.75
−73.00
−73.25
−73.50
−73.75
−74.00
Dec (deg)
169
170
171
172
173
174
RA (deg)